

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.2.2: Multimedia: Sound

Concept 1.2.2.1: Sound Representation and Encoding (Sampling, Rate, Resolution)

Total Questions: 7

Mark Schemes begin on Page 5

Question 1 | Source: 9618_s25_qp_13 (Q1.b)

(b) A teacher records an audio file of the student playing the piano during a music lesson.

Complete the table by giving the term for each description about sound representation on a computer.

Description	Term
the number of times the amplitude is measured per time interval	
the number of bits used to store each amplitude measurement	
the type of sound wave before it is recorded by a computer	

[3]

Question 2 | Source: 9618_w24_qp_13 (Q6.c.i)

6 A computer system stores text, images and sound.

(c) (i) Explain how an analogue sound wave is converted into digital data.

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..... [2]

Question 3 | Source: 9618_w23_qp_13 (Q1.a)

1 (a) State what is meant by **analogue data**.

..... [1]

Question 4 | Source: 9618_w23_qp_13 (Q1.b)

(b) Draw **one** line from each term to its most appropriate description.

Term	Description
sampling	the number of samples taken per second
sampling rate	taking measurements at regular intervals and storing the values
sampling resolution	the number of bits used to store each sample

[2]

Question 5 | Source: 9618_s23_qp_13 (Q3.c.i)

3 A mobile telephone is used to record a video.

(c) The video includes a sound recording.

(i) Describe how sound is represented in a computer.

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..... [3]

Question 6 | Source: 9618_w22_qp_11 (Q1.d.i)

(d) Sound samples are recorded and saved in a file.

(i) State what is meant by **sampling rate**.

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..... [1]

Question 7 | Source: 9618_w21_qp_12 (Q3.a)

3 Andy likes to play computer games.

- (a) Andy uses several input devices to play the games. These include a keyboard and a microphone.

Describe the principal operation of a microphone.

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..... [3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_13 (Q1.b)

1(b)	1 mark for each correct term:	3								
<table><tr><th>Description</th><th>Sound term</th></tr><tr><td>the number of times the amplitude is measured per time interval</td><td>Sampling rate</td></tr><tr><td>the number of bits used to store each amplitude measurement</td><td>Sampling resolution</td></tr><tr><td>the type of sound wave before it is recorded by a computer</td><td>Analogue</td></tr></table>		Description	Sound term	the number of times the amplitude is measured per time interval	Sampling rate	the number of bits used to store each amplitude measurement	Sampling resolution	the type of sound wave before it is recorded by a computer	Analogue	
Description	Sound term									
the number of times the amplitude is measured per time interval	Sampling rate									
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the type of sound wave before it is recorded by a computer	Analogue									

Mark Scheme for Question 2 | Source: 9618_w24_ms_13 (Q6.c.i)

6(c)(i)	1 mark for each bullet point (max 2) - Value/magnitude/size of the analogue sound wave is measured a set number of times each second/time / at set intervals - Each sample/reading/measurement is given the binary number and stored in sequence	2
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Mark Scheme for Question 3 | Source: 9618_w23_ms_13 (Q1.a)

1(a)	1 mark for: data values that are continuously changing // variable // any value	1
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Mark Scheme for Question 4 | Source: 9618_w23_ms_13 (Q1.b)

1(b)

1 mark for 1 correct line
2 marks for 3 correct lines

Term	Description
sampling	the number of samples taken per second
sampling rate	taking measurements at regular intervals and storing the values
sampling resolution	the number of bits used to store each sample

2

Mark Scheme for Question 5 | Source: 9618_s23_ms_13 (Q3.c.i)

3(c)(i)	1 mark each - The amplitude is recorded a set number of times a second - Each (instance of an) amplitude is given a corresponding binary number - The binary number (of each amplitude) is saved in sequence	3
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Mark Scheme for Question 6 | Source: 9618_w22_ms_11 (Q1.d.i)

1(d)(i)	the number of samples taken per unit time / per second	1
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Mark Scheme for Question 7 | Source: 9618_w21_ms_12 (Q3.a)

3(a)	1 mark for each bullet point to max 3 - The microphone has a diaphragm / ribbon - The incoming sound waves cause vibrations of the diaphragm ... causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone) // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) - An electrical signal is produced	3
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